Mechanism Questions #2

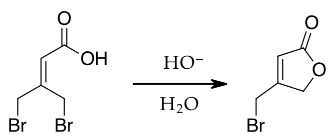
Barry Linkletter

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*.

Explain the reaction mechanisms for the transformations below. Draw out each step and show electron movement using curved arrows. Identify the likely rate-determining step and identify the driving force for each reaction.

Each reaction is a single step in a multistep synthesis described in the referenced literature. For extra experience, access each paper and describe every step in the synthesis that is presented.

1.



"Stereocontrol in the intramolecular Diels-Alder reaction. 1. An application to the total synthesis of (±) marasmic acid." Robert K. Boeckman Jr., Soo Sung Ko, *J. Am. Chem. Soc.*, **1980**, *102*, 7146-7149. https://doi.org/10.1021/ja00543a063

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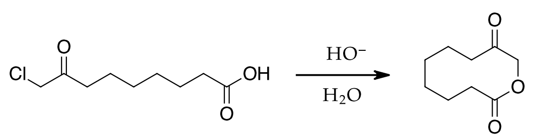
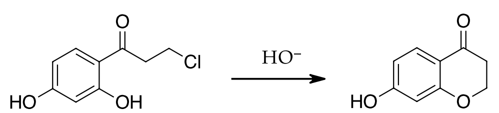


"A new and efficient approach to macrocyclic keto lactones." Mohammad R. Karim, Paul Sampson, *J. Org. Chem.*, **1990**, *55*, 598-605. https://doi.org/10.1021/jo00289a040

3.



"General Preparation of 7-Substituted 4-Chromanones: Synthesis of a Potent Aldose Reductase Inhibitor." Kevin Koch, Michael S. Biggers, *J. Org. Chem.*, **1994**, *59*, 1216-1218. https://doi.org/10.1021/jo00084a050

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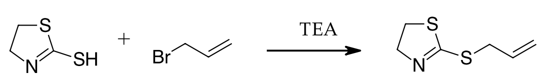

"New approaches to the pyrrolizidine ring system: total synthesis of (±)-isoretronecanol and (±)-trachelanthamidine." Harold W. Pinnick, Yeong-Ho Chang, *J. Org. Chem.*, **1978**, *43*, 4662-4663. https://doi.org/10.1021/jo00418a031

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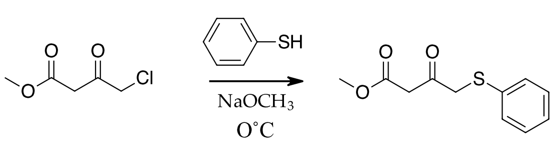
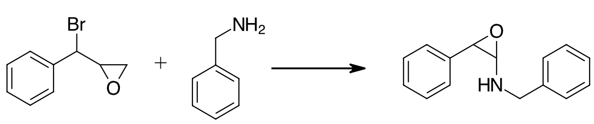


"Synthesis of Phenylthiomethyl Substitution Furans by Lewis Acid Catalysed Substitution." Nicholas Greeves, Julie S. Torode, *Synthesis*, **1993**, 1109-1112. https://doi.org/10.1055/s-1993-26011

6.



"Synthesis of 2-Aminomethyloxiranes." Karikomi Michinori, Yamazaki Tohru, Toda Takashi, *Chemistry Letters*, **1993**, *22*, 1787-1790. https://doi.org/10.1246/cl.1993.1787.

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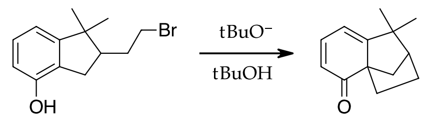
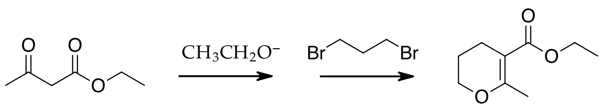


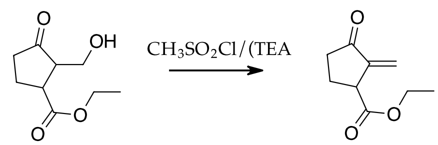
"Aryl participated cyclisations involving indane derivatives a total synthesis of (±)-isolongifolene." Swati Das, Tapan Kr. Karpha, Manuka Ghosal, Debabrata Mukherjee, *Tetrahedron Letters*, **1992**, *33*, 1229–1232. https://doi.org/10.1016/S0040-4039(00)91904-X Available in UPEI Library stacks.

8.



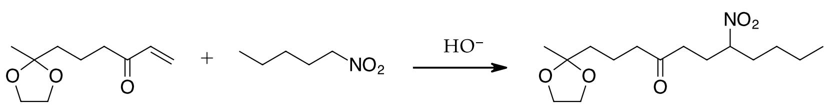
"Ring-size effect in the reaction of α,ω-dibromoalkanes with ethyl acetoacetate and its application to the synthesis of some disubstituted oxa- and aza-heterocycles." Lin Yong-Yue, Liang Xiao-Guang (Hiu-Kuang Leung), Chan Yuk-Yee, *Chin. J. Chem.*, **1990**, *8*, 153–159. https://doi.org/10.1002/cjoc.19900080211. Available via ILL.

9.



"The INOC route to carbocyclics: a formal total synthesis of (±)-sarkomycin." Alan P. Kozikowski, Philip D. Stein, *J. Am. Chem. Soc.*, **1982**, *104*, 4023–4024. https://doi.org/10.1021/ja00378a049

10.



"Studies on the stereochemistry of nucleophilic additions to tetrahydropyridinium salts. A stereospecific total synthesis of (±)-monomorine I." Robert V. Stevens, Albert W. M. Lee, *J. Chem. Soc., Chem. Commun.*, **1982**, 102–103. https://doi.org/10.1039/C39820000102

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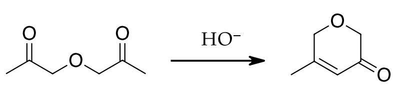
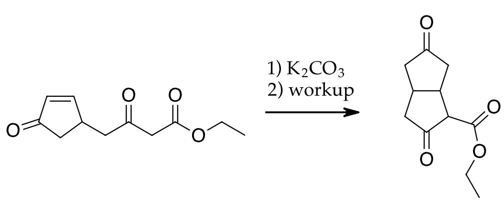


"Synthesis of 2H-Pyran-3-(6H)-ones." Kari Skinnemoen, Kjell Undheim, *Acta Chemica Scandinavica B*, **1980**, *34*, 295–297. https://doi.org/10.3891/acta.chem.scand.34b-0295

12.



"A new, elegant route to a key intermediate for the synthesis of 9(0)-methanoprostacyclin." Achille Barco, Simonetta Benetti, Gian Piero Pollini, Pier Giovanni Baraldi, Carmelo Gandolfi, *J. Org. Chem.*, **1980**, *45*, 4776–4778. https://doi.org/10.1021/jo01311a047

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will have…

- reviewed organic chemistry mechanisms involving basic conditions.